

# Choosing the $B(c)$ correlation for identifiable fits and contour seeding

Why  $B$  and  $\sigma$  hit the walls — and the solute-permeability equation that fixes the contours, the seeds, and the fits.

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Follow-up to the June 11 collaborator meeting • June 2026

# 01

## Where we left off

*The June 11 result — and the problem it left open*

# Identifiability degraded with cation valence

Forward-simulated WSSE contours over  $\theta = (L_p, B, \sigma)$  sorted 11 single-salt sheets into three stages

## STAGE 1 • WELL-IDENTIFIED

2 sheets • NaCl

Sharp interior minimum.  $L_p$  pinned;  $\sigma$  leaves a real fingerprint. Trustworthy fit.

fit: GOOD

## STAGE 2 • INCONSISTENT

7 sheets • NaCl + CaCl<sub>2</sub>

$L_p$  tight, but  $\sigma$  (often B) pinned at a wall — channels minimise at different walls.

fit: OKAY

## STAGE 3 • NOT ESTIMABLE

2 sheets • LaCl<sub>3</sub>

No coherent basin; WSSE ~20× the NaCl reference. Constant-( $\sigma, B$ ) model cannot fit.

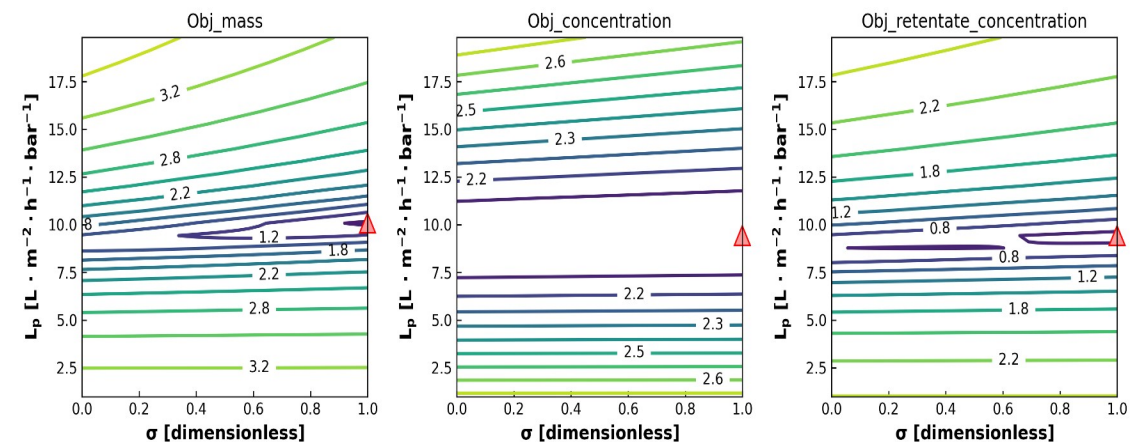
fit: POOR

**Monotonic trend:** each added cation charge pushed the fit from a clean interior optimum toward the bounds.

# B and $\sigma$ were unidentifiable or pinned at the bounds

Across most single-salt sheets the optimizer landed on a wall, not an interior optimum

- $\sigma \rightarrow 1$  wall — On most NaCl / CaCl<sub>2</sub> sheets  $\sigma$  railed to its upper bound.
- $B \rightarrow$  bound — B floated to 0 or to its per-salt ceiling (e.g. 30  $\mu\text{m/s}$ ).
- Valley, not a point — Channels each found a minimum — at different ( $\sigma$ ,  $L_p$ ) walls.
- Worse with valence — LaCl<sub>3</sub>: no stationary basin under either parameter sweep.



$\sigma \times L_p$  WSSE contour — interior optimum (left) vs pinned at the  $\sigma = 1$  wall (right)

The question this raised: *is this a data-geometry problem, or are we fitting the wrong B?*

# 02

## Why — data geometry, or model form?

*Reading the walls as a symptom of a mis-specified  $B(c)$*

# A wall is what a mis-specified $B(c)$ looks like

*If  $B$  truly varies across a run but the model holds it constant/linear,  $\sigma$  and  $B$  absorb the mismatch*

Within one diafiltration run the feed-side concentration sweeps over a wide range, so the true  $B$  changes during the run. A constant or single-slope  $B$  cannot follow it — the optimizer compensates by driving  $\sigma$  to 1 and  $B$  to a bound. The fix is not more freedom in  $\sigma$ ; it is the right functional form for  $B(c)$ .

## Symptom

- $\sigma$  pinned at 1
- $B$  at 0 or ceiling
- channels disagree

## Usual reading

- “ $\sigma$ ,  $B$  unidentifiable”
- add data / constraints
- re-weight channels

## This work's reading

- $B(c)$  form is wrong
- tie  $B$  to concentration
- walls relax

# The same B across all three papers

*Solute permeability = membrane diffusion × partition, per unit thickness — a definition, not a new model*

$$B = D_m \cdot H / \ell = P_0 \cdot K$$

$$J_s = B (c_{in} - c_H)$$

$D^\oplus$  membrane diffusivity ·  $H$  partition coefficient ·  $\ell$  active-layer thickness ·  $P_0 = D^\oplus / \ell$  ·  $K$  = salt partition

DATA1 / DATA2

$$B = D^\oplus H / \ell$$

Transport SI (collaborators)

$$B_{ii} = D_i H_i / \ell$$

This work (global)

$$B = P_0 \cdot K, K \text{ from partition}$$

*Same symbol, three notations. All the concentration dependence lives in the partition  $H$ .*

03

## The right equation for $B(c)$

*From the shape of the partition to the form you fit*

# B(c) is a saturating partition — Donnan + dielectric

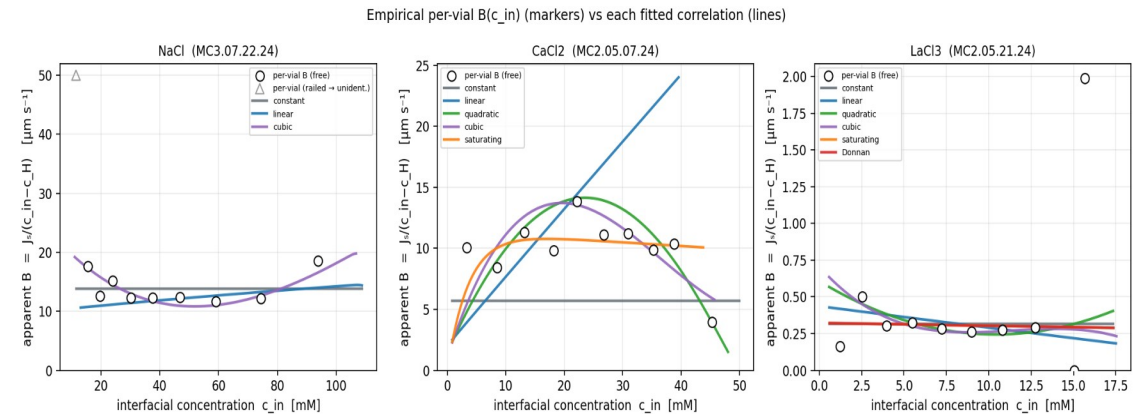
Near-zero at infinite dilution, steep through the middle, plateau once the fixed charge is screened

$$B(c) = P_0 [-X + \sqrt{(X^2 + 4k^2c^2)}] / (2c)$$

$$c \rightarrow 0: B \approx (P_0 k^2 / X) c \quad c \rightarrow \infty: B \rightarrow P_0 k$$

**Two physical knobs:**  $P_0 \cdot k$  sets the high-salt plateau,  $X$  sets the concentration of the knee where Donnan exclusion gives way.

Monotonic and concave — the opposite curvature of a growth exponential.



Mechanistic B(c): saturating, not exponential, not an unbounded polynomial

# Inside the window: a low-order Taylor. Across it: saturating.

Campaign DAE fits across the single-salt sheets — in-window from per-sheet WSSE, extrapolation from the fitted partition curves

| Form                | in-window    | extrapolation     | Verdict     |
|---------------------|--------------|-------------------|-------------|
| Linear (order 1)    | good         | unbounded ↑       | interp.     |
| Quadratic (order 2) | better       | $B < 0$ (unphys.) | interp.     |
| Cubic (order 3)     | best in-win. | diverges          | interp.     |
| Saturating exp      | near-best    | bounded ✓         | robust      |
| Mechanistic Donnan  | near-best    | bounded ✓         | mechanistic |

$$B = Jw \cdot \sum \beta_i c_{in}^i$$

$$B = B_{\infty} (1 - e^{(-c/c^*)})$$

On DATA3 (per-sheet WSSE):  $B(c)$  matters most for  $\text{CaCl}_2$  — quadratic / cubic / saturating all cut the error  $\approx 33\%$  vs constant  $B$  ( $220 \rightarrow 146$ ). Higher-order Taylor edges the in-window fit but extrapolates to unphysical  $B$  (the quadratic bends to  $B < 0$ ); saturating matches it and stays bounded — the practical form.  $\text{NaCl}$  is fine with  $\sim$ constant  $B$ ;  $\text{LaCl}_3$  stays poor under every  $B(c)$ , pointing past  $B(c)$  alone.

Don't extrapolate it — a negative  $c^2$  term bends  $B$  downward, which is unphysical.

For transfer across the campaign — fit the saturating / mechanistic form.

# The DATA2 linear B is the first-order Taylor of this curve

*Their clean correlation and the saturating partition are the same object — local vs global*

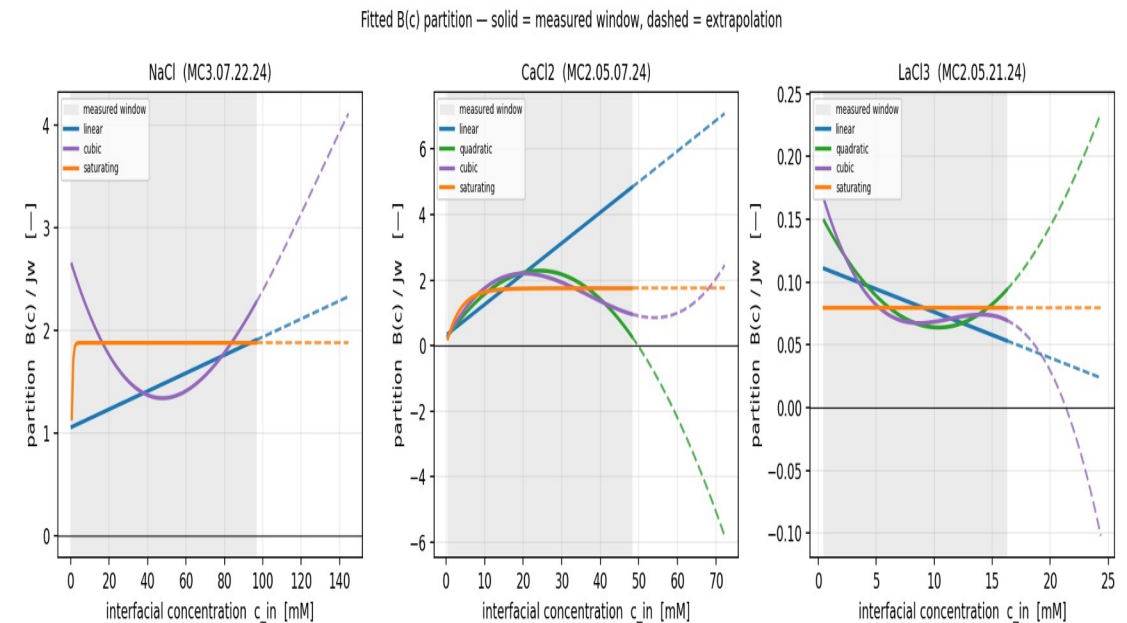
$$B \approx \beta_0 + \beta_1 c, \quad \beta_1 = dB/dc$$

DATA2 fits  $B = Jw \cdot \sum \beta_i c^i$  — a truncated Taylor series of the partition  $H(c)$ .

Matched to their 15–120 mM window, the fitted slope  $\beta_1 = 0.248$  equals the curve's derivative  $dB/dc = 0.250$ .

Their AIC pick (linear) is reproduced: with realistic noise, linear wins 82% / quadratic 12%.

**Local polynomial and global mechanistic are two faces of one  $B(c)$ .**



*Linear fit hugs the curve in-window, peels off outside*

# Higher valence sharpens the knee and lowers the plateau

Exactly the  $\text{LaCl}_3$  Stage-3 result — a model-form limit, not just thin data

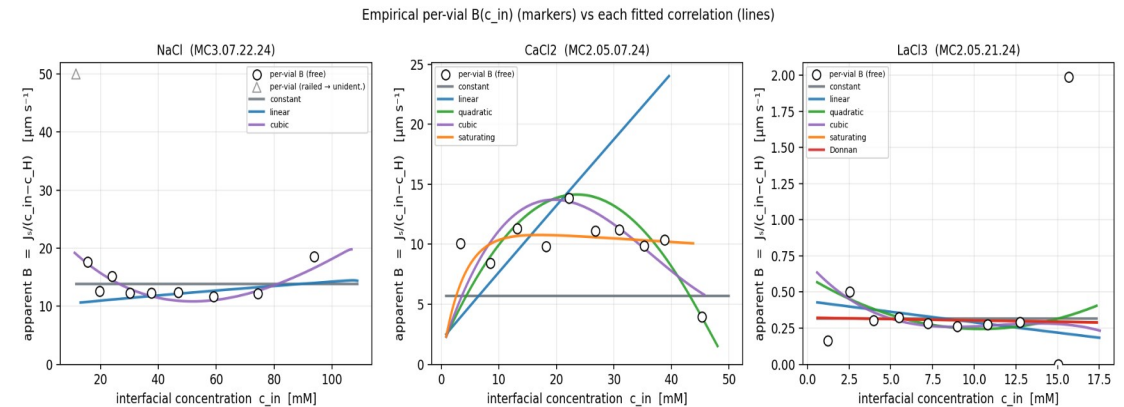
$$B \propto c^{(1/z)}: c \text{ (1:1)}, \sqrt{c} \text{ (2:1)}, \sqrt[3]{c} \text{ (3:1)}$$

The low- $c$  rise steepens from  $c$  (1:1) to  $\sqrt{c}$  (2:1) to  $\sqrt[3]{c}$  (3:1) — a near-vertical knee.

A constant or linear  $B$  cannot represent that knee, so  $\text{CaCl}_2$  rails  $\sigma$  and  $\text{LaCl}_3$  shows no basin.

Dielectric (Born) exclusion also lowers the plateau as charge rises — your fitted  $\beta_0$  ordering  $\text{NaCl} \gg \text{CaCl}_2 \gg \text{LaCl}_3$  ( $\propto e^{(-0.32 z^2)}$ ).

**So Stage 3 is the model talking, not the data failing.**



Knee sharpens with  $z$ ; a single curvature can't fit all three

# Single-salt: a relabel. Multisalt: the only common axis.

*Ionic strength is what sets Donnan screening — keep it ready for the mixtures*

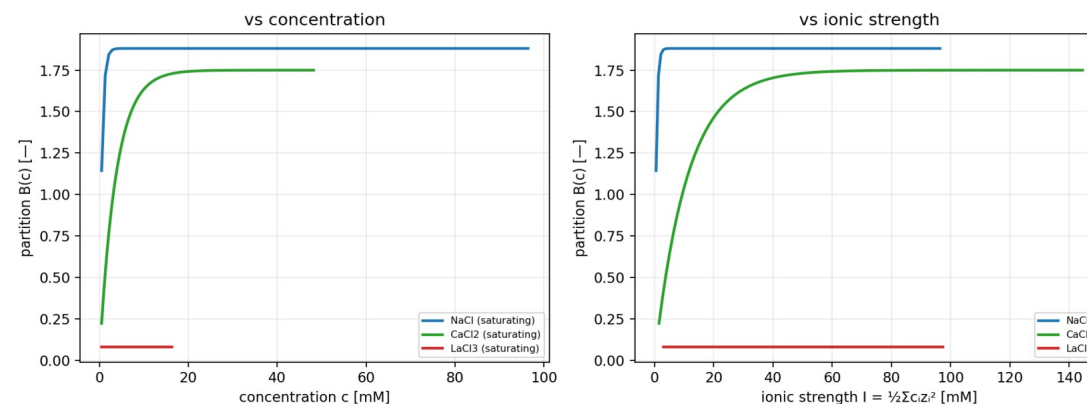
$$I = \frac{1}{2} \sum c_i z_i^2 = c_{\text{Na}} + 3 c_{\text{Ca}} + 6 c_{\text{La}}$$

For one salt  $I \propto c$ , so B-vs-I is B-vs-c relabeled — no change to a single-salt fit.

Switching the axis to I pulls the salts ~3× closer together (partial collapse).

**For a mixture there is no single c — I is the only well-defined abscissa.**

*Single-salt focus now; this is the bridge to the multisalt phase.*



Same curves on a c-axis (left) vs an I-axis (right): tighter on I

# 04

## Using $B(c)$ in the contours

*How the right form fixes seeding and identifiability*

# Tie B to concentration and it stops floating to a wall

One physical function replaces a free B per range or per vial — fewer parameters, sharper basins

## Before — free constant/linear B

B is unconstrained across the run, so it slides to a bound and drags  $\sigma$  with it.

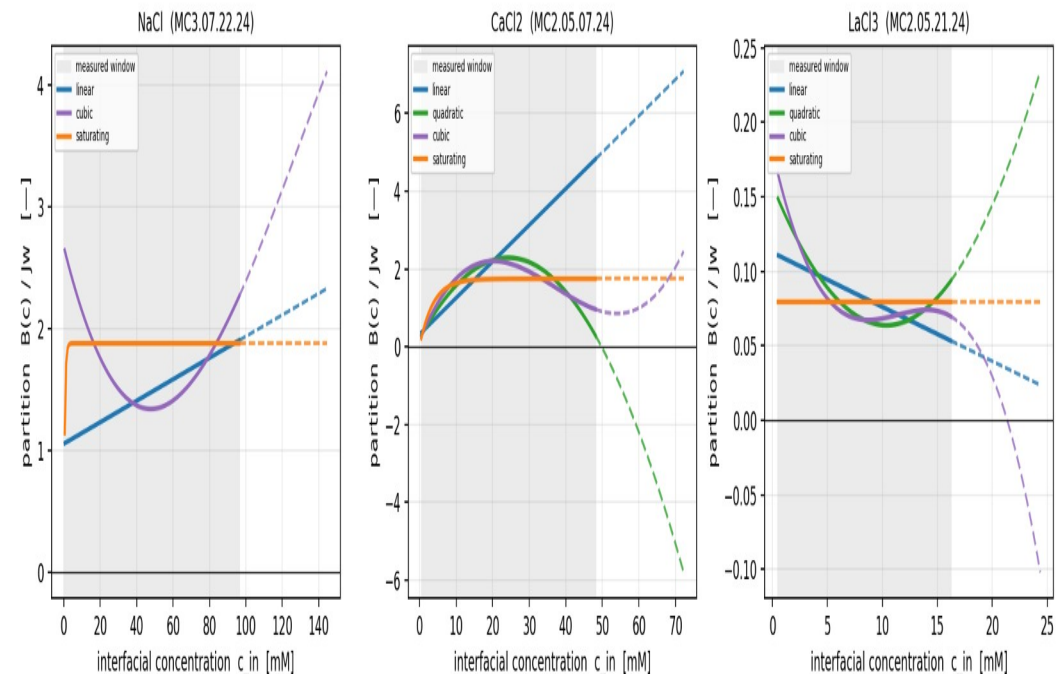
## After — $B = P_0 \cdot K(c)$

B is fixed by concentration through one physical curve; only  $P_0$ ,  $X(k)$  and  $\sigma$  remain free.

## Initial guesses

The “group by range / by vial” question becomes one  $B(c)$  — basins from different ranges line up.

Fitted  $B(c)$  partition — solid = measured window, dashed = extrapolation



Local (linear) vs global (saturating)  $B(c)$  for one salt

# Read the seed off the contour, then fit

WSSE contour (forward-sim, no fitting) → joint argmin → seed  $\theta$  → optimize

1

Map

Sweep the grid; score WSSE per channel

2

Read

Joint 2-D minimum where channels agree

3

Seed

Use that  $\theta$  as the initial guess

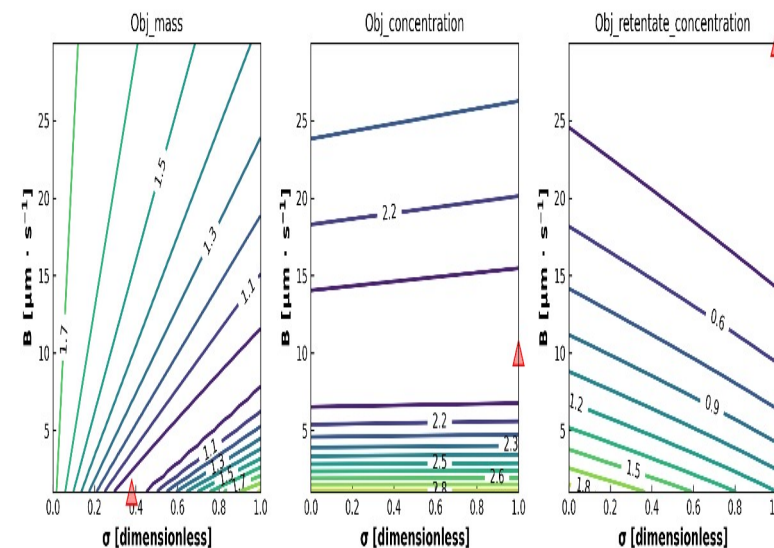
4

Fit

Optimize with  $B = B(c)$  fixed by the curve

$$WSSE = \sum_{ch} \sum_t w (y - \hat{y})^2$$

Forward simulation, no fitting. Where the channels agree on an interior minimum,  $\theta$  is identified — and that point seeds the optimizer.



Read the joint minimum straight off the contour

# From “guess per range / per vial” to one $B(c)$

*The grouping question answers itself once  $B$  is tied to a physical variable*

## Per-vial free $B$

many parameters, overfits, no transfer

## Grouped by range

linearizes a curve the polynomial should already capture

## $B(c)$ — one function

ties every vial's  $B$  to its concentration; multistart seeds from the curve

*Keep LHS multistart — but seed it from the contour argmin and the  $B(c)$  curve, not a blind start.*

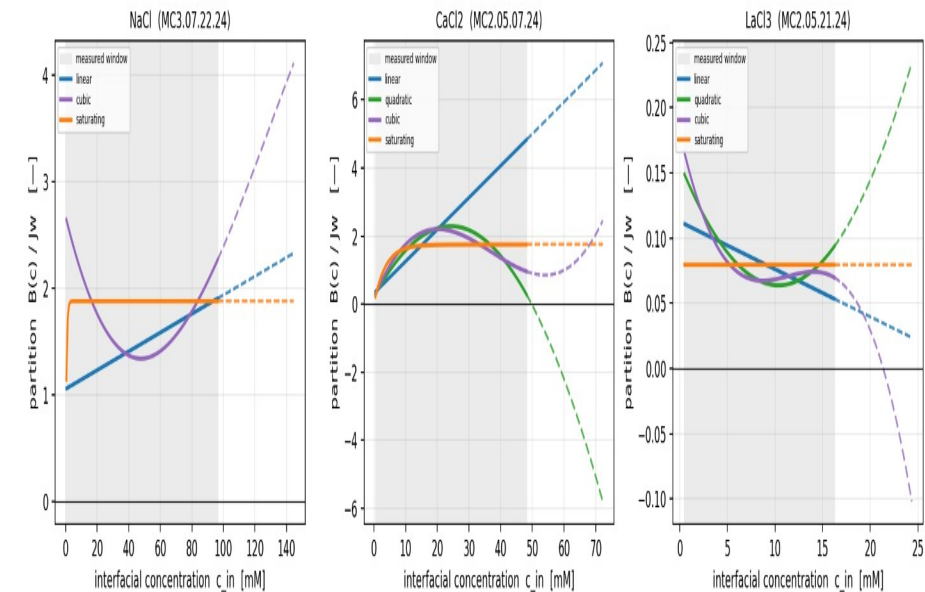
# Why we don't map each $\beta$ against $L_p$

The  $\beta$ 's are coupled pieces of one curve — not independent dials

- **One curve, not three knobs** —  $B(c) = Jw(\beta_0 + \beta_1 c + \beta_2 c^2)$  is a single curve; the  $\beta$ 's are its pieces.
- **They lean on each other** — Raise  $\beta_1$  and the fit pushes  $\beta_0$  and  $\beta_2$  back to keep the curve through the data — so the  $\beta$ 's are correlated.
- **So the slice is an artifact** — An  $L_p \times \beta_1$  map must freeze  $\beta_0$  and  $\beta_2$ . What you see depends on where you froze them — not on  $\beta_1$ .
- **$L_p$  is the clean one** —  $L_p$  is set by the water flux and barely moves with the  $\beta$ 's — so every  $L_p \times \beta$  looks the same:  $L_p$  pinned.

**Analogy:** judging one singer with the rest of the choir muted — you mostly hear your choice of who to mute.

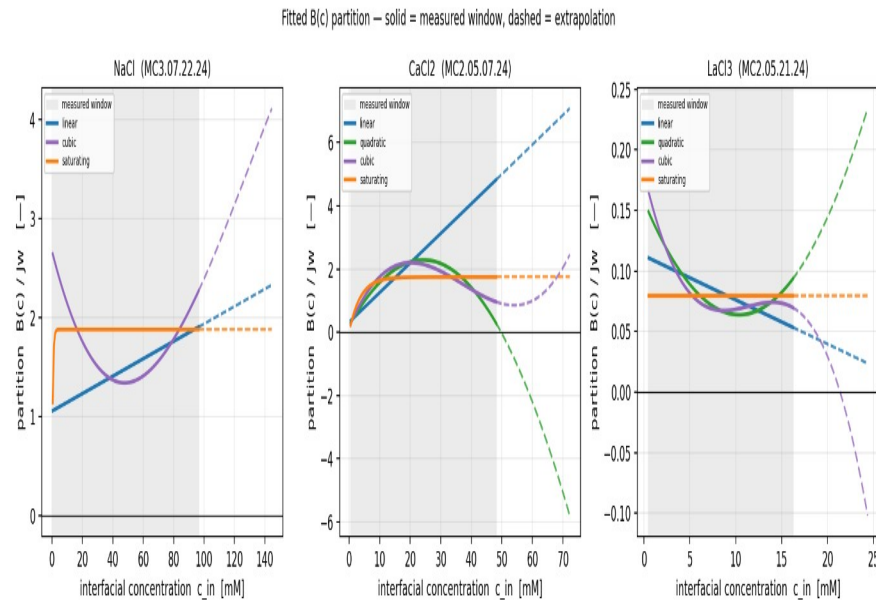
Fitted  $B(c)$  partition — solid = measured window, dashed = extrapolation



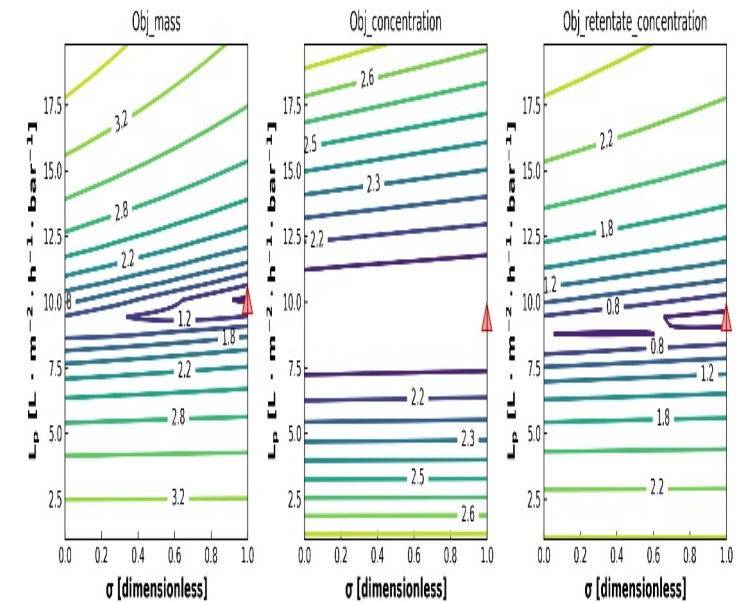
$\beta_1 \times \beta_2$  : a long valley, no single point — that's the correlation

# What to plot for the $\beta$ 's instead

Two views carry all the  $\beta$  information — no per- $\beta$  maps needed



The  $B(c)$  curve + per-vial apparent  $B$  that constrains it (illustrative)



FIM correlation matrix — identifiability &  $\beta$  coupling, exact

**Full set:** keep  $\sigma \times L_p$ ,  $B \times L_p$ ,  $B \times \sigma$  for  $L_p$  and  $\sigma$ ; these two cover the  $B$ -function. The  $\beta$ 's live on the curve and in the matrix — never in one- $\beta$ -at-a-time slices.

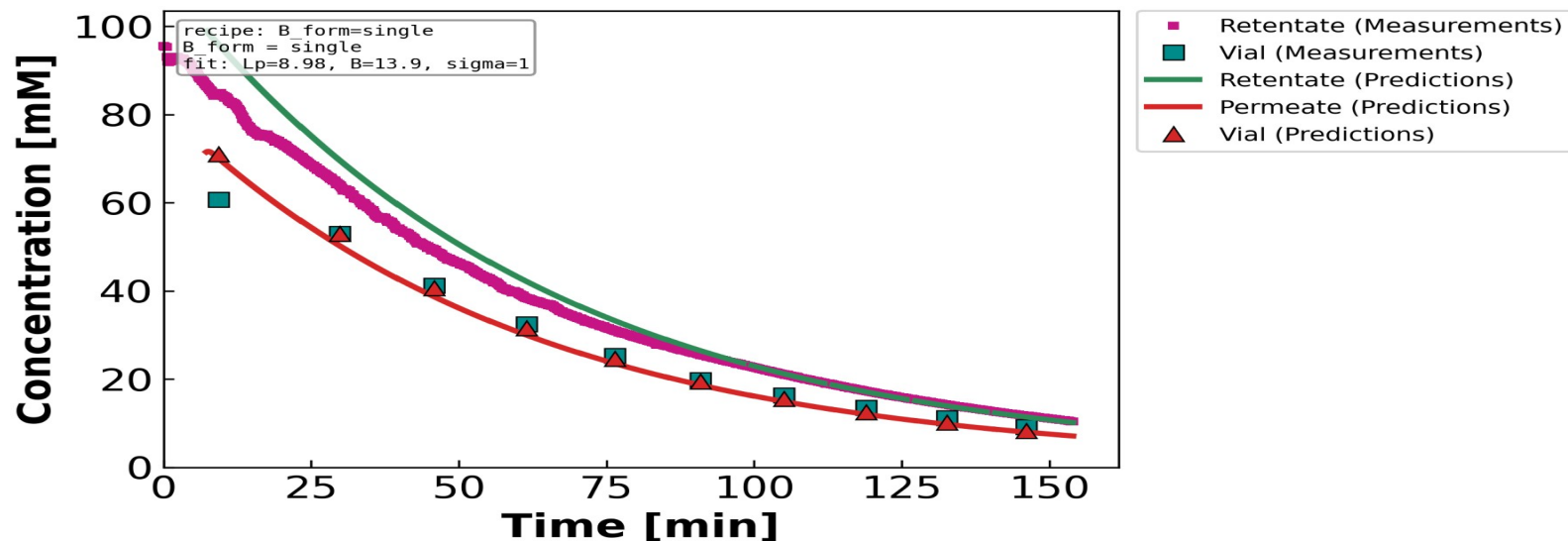
05

## Predictions vs measured states

*Mass and concentration, model against experiment*

# NaCl: model tracks mass and both concentrations

MC3 • 07.22.24 • SNaCl — constant-B fit ( $B_{\text{form}} = \text{single}$ )



● experiment — model • mass (left) and retentate/permeate concentration (right)

$\theta$  (contour seed)

$L_p$  8.98

$B$  13.88  $\mu\text{m/s}$

$\sigma$  1.00

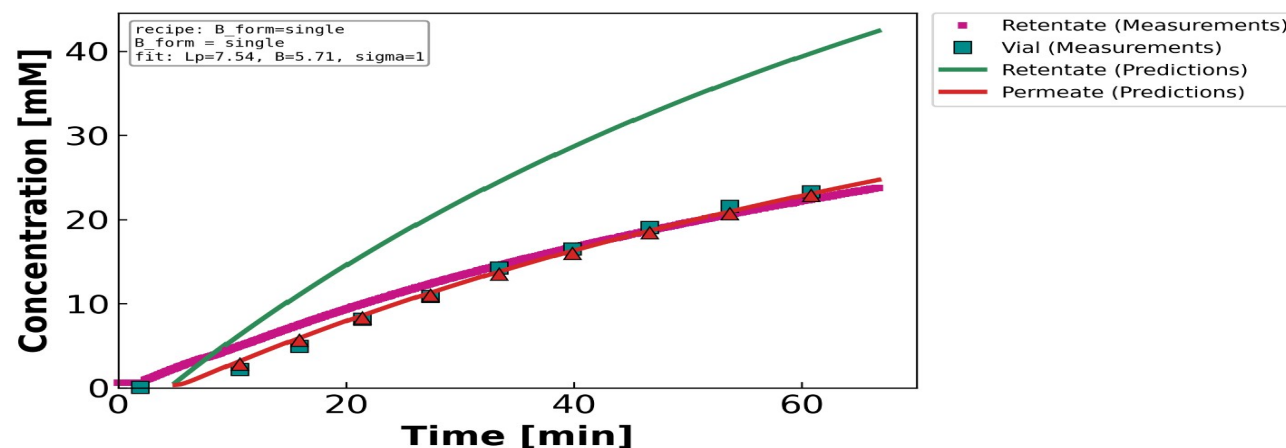
WSSE<sub>3ch</sub> 42

GOOD

$B$  is ~constant for NaCl over this range — the constant-B fit already tracks mass and both concentrations (the per-vial  $B$  is flat, slide 9).

# Divalent fits with $\sigma$ railed; trivalent the model can't track

The case for  $B(c)$ : the constant- $(\sigma, B)$  form runs out as valence rises



$\text{CaCl}_2$  • MC2 • 05.07.24

$L_p$  7.54

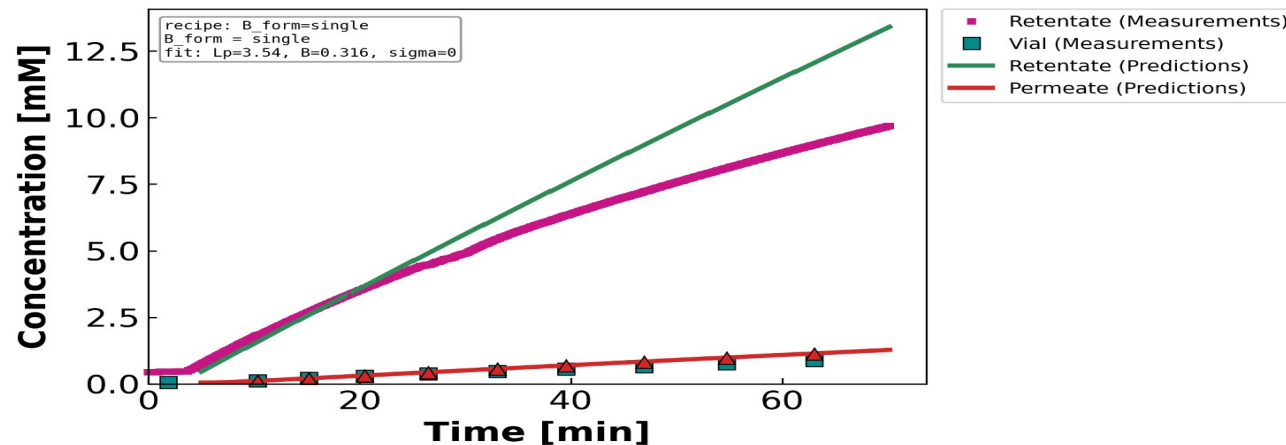
$B$  5.71  $\mu\text{m/s}$

$\sigma$  1.00  $\uparrow$

$\text{WSSE}_3\text{ch}$  220

OKAY

$\sigma$  railed at 1; the saturating  $B(c)$  cuts  $\text{WSSE}$  220  $\rightarrow$  146 (slide 10).



$\text{LaCl}_3$  • MC2 • 05.21.24

$L_p$  3.54

$B$  0.32  $\mu\text{m/s}$

$\sigma$  0.00  $\downarrow$

$\text{WSSE}_3\text{ch}$  1244

POOR

Model-limited — every  $B(c)$  form stays  $\sim$ 1240;  $B(c)$  alone can't rescue it.

06

## Takeaways & next steps

*What to change before the re-run*

# The B(c) form is the lever for identifiable fits

## B is a saturating partition

$B = D^{\oplus}H/\ell$ ; H rises then plateaus (Donnan + dielectric).

## Pick by purpose

In-window: Taylor B\_form (1-2). Across the campaign: saturating / mechanistic.

## Valence = model form

Sharper knee + lower plateau with z explains  $\text{CaCl}_2$  walls and  $\text{LaCl}_3$ .

## Seed from B(c)

Tie B to concentration → walls relax, contours sharpen, seeds transfer.

## NEXT STEPS

**Now (single-salt):** re-fit with the chosen B\_form seeded from the contours. **Soon (multisalt):** move to B(I) on the shared ionic-strength axis.